

be the agent

artificial intelligence · prof. dr. nicolas meseth

1 the game

2 play

3 what you just played

today, you
are the agent,
and the computer
is your group.



part 1

the game

seven roles

model

sees only its stack of cards

seven roles

search

web

files

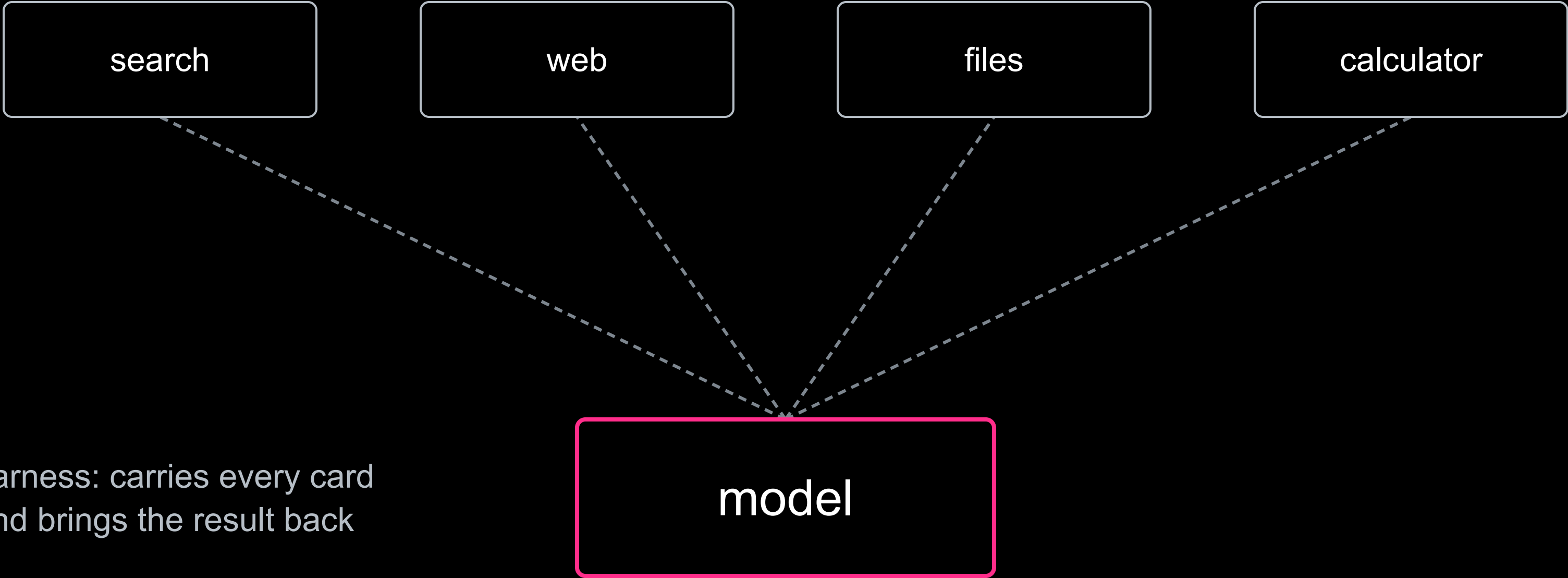
calculator

model

sees only its stack of cards

the game

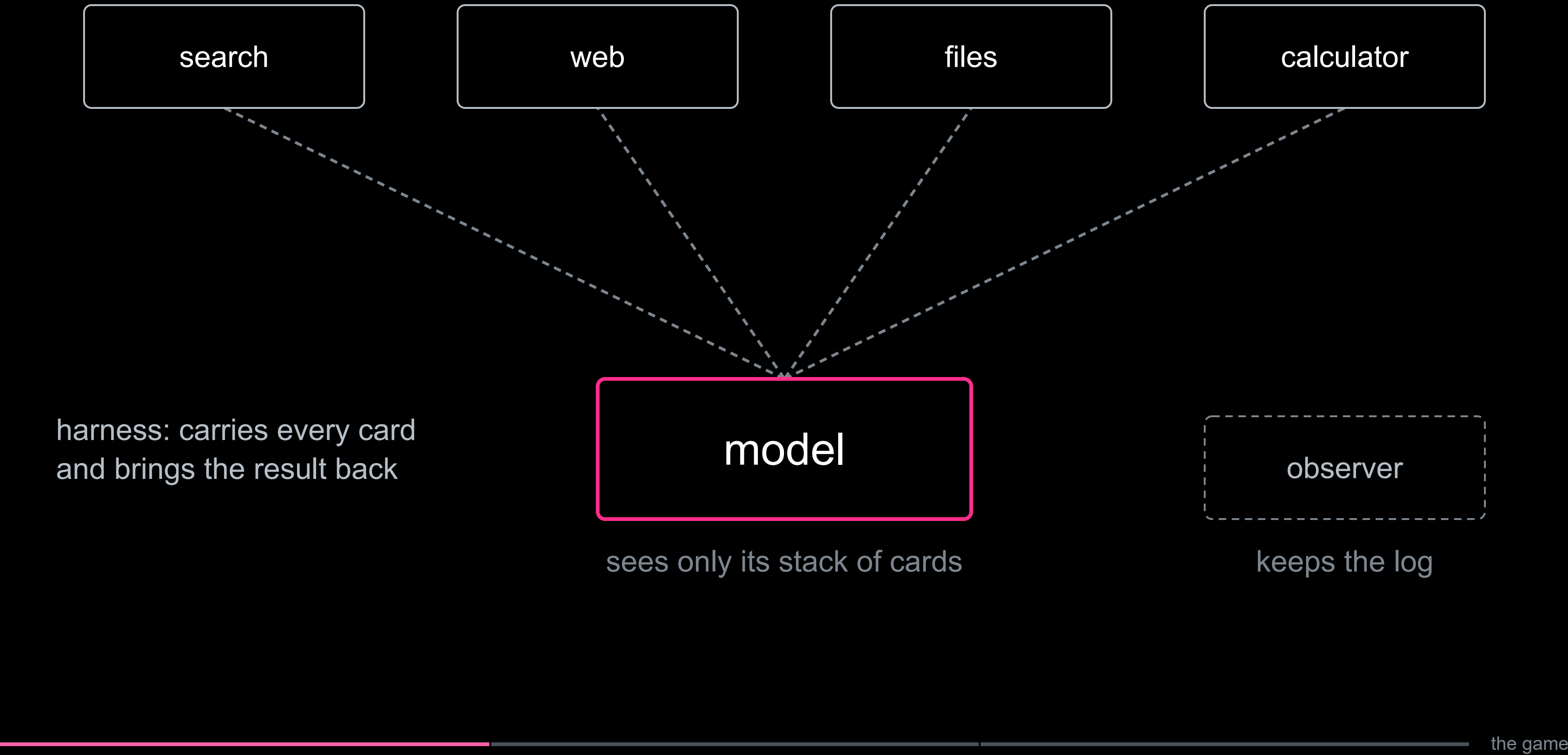
seven roles



harness: carries every card
and brings the result back

sees only its stack of cards

seven roles



four kinds of cards

THINK :

i need the size
of the field first.

four kinds of cards

THINK :

i need the size
of the field first.

TOOL :

```
read_file(  
"fields")
```

four kinds of cards

THINK :

i need the size
of the field first.

TOOL :

```
read_file(  
"fields")
```

RESULT :

long acre,
4.5 ha,
winter wheat

four kinds of cards

THINK :

i need the size
of the field first.

TOOL :

```
read_file(  
"fields")
```

RESULT :

long acre,
4.5 ha,
winter wheat

ANSWER :

4.5 ha,
winter wheat

the rules

the model sees only its stack, and nobody talks to it.

the game

the rules

the model sees only its stack, and nobody talks to it.

the harness carries every card to the right tool and brings the result back.

the model sees only its stack, and nobody talks to it.

the harness carries every card to the right tool and brings the result back.

a tool does exactly what the card says, nothing more.

the rules

the model sees only its stack, and nobody talks to it.

the harness carries every card to the right tool and brings the result back.

a tool does exactly what the card says, nothing more.

one card at a time, until the model writes ANSWER :.

the system card

<code>search("words")</code>	finds web pages and returns their titles
<code>open("title")</code>	returns the full text of one web page
<code>read_file("name")</code>	returns a farm file: fields, soil_tests, stock, weather
<code>calculator("expression")</code>	calculates, e.g. <code>calculator("4.5 * 100")</code>
<code>write_file("name", "text")</code>	saves a text as a new farm file

this is everything the model knows about its tools.

part 2

play

task 1 · about 5 minutes

how large is the field long acre, and what grows on it this year?

task 2 · swap roles · about 12 minutes

how much nitrogen may we still apply to long acre this spring? give the total in kg N for the whole field.

which nitrogen requirement did your model use, and where did it come from?

task 3 · if there is time

plan the first nitrogen application on long acre: how much of which product do we need, and can we spread it on 10 february?

part 3

what you just played

from the game to a real agent

the stack of cards → **context window**: everything the model knows

from the game to a real agent

the stack of cards → **context window**: everything the model knows

the system card → **system prompt**: who it is, which tools it has

from the game to a real agent

the stack of cards → **context window**: everything the model knows

the system card → **system prompt**: who it is, which tools it has

THINK : → **reasoning**, the plan

from the game to a real agent

the stack of cards → **context window**: everything the model knows

the system card → **system prompt**: who it is, which tools it has

THINK : → **reasoning**, the plan

TOOL : → **tool call**, written as text

from the game to a real agent

the stack of cards → **context window**: everything the model knows

the system card → **system prompt**: who it is, which tools it has

THINK : → **reasoning**, the plan

TOOL : → **tool call**, written as text

the harness → **harness**: the program that runs the tools

from the game to a real agent

the stack of cards → **context window**: everything the model knows

the system card → **system prompt**: who it is, which tools it has

THINK : → **reasoning**, the plan

TOOL : → **tool call**, written as text

the harness → **harness**: the program that runs the tools

RESULT : → **observation**, added to the context

from the game to a real agent

the stack of cards	→	context window : everything the model knows
the system card	→	system prompt : who it is, which tools it has
THINK :	→	reasoning , the plan
TOOL :	→	tool call , written as text
the harness	→	harness : the program that runs the tools
RESULT :	→	observation , added to the context
ANSWER :	→	stop : the goal is reached

the same, in a real agent

TOOL	list the files in the folder <code>Get-ChildItem -Recurse</code>
RESULT	one file: <code>WS 26_27 - Applications of Artificial Intelligence.md</code>
TOOL	read that file <code>Read("WS 26_27 -md")</code>
RESULT	“a brief history of artificial intelligence ... 1950: alan turing ...”
THINK	“before the plan, two things are missing: there's no pinboard photo, and there's no file called milestones.md.”
TOOL	start eight sub-agents <code>Agent("G1 foundations 1943–1969") ...</code>

an AI agent is a language model that can only write text; everything it does in the world happens through tools that the harness runs for it.

reflect

in which round would the model have been stuck without the tools?

reflect

in which round would the model have been stuck without the tools?

what would have changed if the model had to give up its stack after ten cards?

reflect

in which round would the model have been stuck without the tools?

what would have changed if the model had to give up its stack after ten cards?

who did the thinking, who did the acting, and which of it could the model actually do?